REGULATORY COMPLIANCE ASSESSMENT

21 CFR Part 11

Electronic Records & Electronic Signatures Compliance Assessment

BioVenture Inc. — eTMF & ELN Systems

Phase II Oncology Clinical Trial: BVI-2204

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Report Status	FINAL — For Client Review and Action
Prepared By	Dr. M. Chen, PhD BME / JD — Principal Regulatory Consultant
Systems Assessed	Veeva Vault eTMF (v24.1) | LabArchives ELN (v8.3)
Trial Reference	IND 145-2204 — BVI-101 (anti-PD-L1 / VEGFR2 bispecific) Phase II
Applicable Regulation	21 CFR Part 11; 21 CFR Parts 312 & 314; ICH E6(R3)
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Section 1 — Executive Summary

This 21 CFR Part 11 Compliance Assessment was commissioned by BioVenture Inc. ("BioVenture" or "the Company") in advance of the activation of its Phase II oncology clinical trial (IND 145-2204, Study BVI-2204) evaluating BVI-101, an investigational anti-PD-L1/VEGFR2 bispecific antibody in recurrent or refractory non-small cell lung cancer (NSCLC). The assessment evaluates two systems newly implemented for the trial: the Veeva Vault electronic Trial Master File (eTMF) and the LabArchives Electronic Laboratory Notebook (ELN). Both systems create, modify, maintain, archive, retrieve, or transmit electronic records that are subject to FDA regulations administered under 21 CFR Part 11 and its predicate rules (21 CFR Parts 312 and 314).

The assessment was conducted through a combination of system configuration review, documentation audit, structured interviews with system administrators and end users, and testing of system functionality against the regulatory requirements of 21 CFR §§ 11.10, 11.30, 11.50, 11.70, 11.100, and 11.200. Findings are classified on a four-tier risk scale (Critical / High / Medium / Low) corresponding to the severity of potential regulatory impact and the likelihood of FDA citation in the event of an inspection.

1.1 Overall Compliance Status

The traffic light framework below reflects the aggregate compliance posture for each system and requirement domain assessed. A RED status indicates material non-compliance that poses immediate inspection risk and must be remediated before the system is used for regulated trial activities. YELLOW indicates partial compliance with significant gaps requiring remediation within 90 days. GREEN indicates compliance with at most minor documentation gaps.

Requirement Domain	eTMF (Veeva Vault)	ELN (LabArchives)	Combined Status
System Validation	YELLOW	RED	RED
Audit Trails	GREEN	YELLOW	YELLOW
Access Controls	GREEN	YELLOW	YELLOW
Electronic Signatures	YELLOW	RED	RED
Records Retention & Recovery	GREEN	YELLOW	YELLOW

1.2 Top Three Priority Findings

PRIORITY FINDING 1 — CRITICAL: LabArchives ELN Validation Not Complete (21 CFR § 11.10(a))

The LabArchives ELN has not been formally validated for use in GxP-regulated activities. No Installation Qualification (IQ), Operational Qualification (OQ), or Performance Qualification (PQ) documentation exists. The system is currently being used to record nonclinical pharmacology and toxicology data supporting IND 145-2204. Use of a non-validated computerized system to generate, store, or retrieve electronic records subject to FDA regulation violates 21 CFR § 11.10(a) and constitutes a data integrity risk that could compromise the evidentiary value of nonclinical data in the IND. Immediate remediation required before continued GxP use.

PRIORITY FINDING 2 — CRITICAL: Electronic Signatures in ELN Not 21 CFR Part 11 Compliant (21 CFR §§ 11.50, 11.100, 11.200)

Electronic signatures applied in LabArchives ELN do not meet the requirements of 21 CFR §§ 11.50, 11.100, and 11.200. Specifically: (a) signature manifestations do not display the printed name of the signer, the date/time of signing, or the meaning of the signature (e.g., authorship, review, approval) as required by § 11.50(a); (b) the system has not been configured to use at least two distinct identification components as required by § 11.200(a)(1); and (c) the Company has not submitted the § 11.100(c) certification to FDA confirming that electronic signatures are the legally binding equivalents of handwritten signatures. Records signed electronically in the ELN may not be legally valid for FDA submission purposes.

PRIORITY FINDING 3 — HIGH: Veeva Vault eTMF Electronic Signature Manifestation Non-Compliant; Periodic Audit Trail Review SOP Absent (21 CFR §§ 11.10(e), 11.50)

The Veeva Vault eTMF produces compliant electronic signatures for initial document approval workflows; however, the signature manifestation for amendment/revision approvals does not display the meaning of the signature as required by § 11.50(a)(3). Additionally, no Standard Operating Procedure (SOP) governs the periodic review of eTMF audit trails, as required by FDA expectations articulated in the 2017 FDA Data Integrity Guidance. Audit trail review has not been performed since system go-live six months ago. These gaps, taken together, indicate a systemic audit trail governance deficiency that will be cited in a routine FDA clinical inspection.

1.3 Assessment Conclusion

BioVenture Inc. faces material 21 CFR Part 11 compliance gaps in both systems. The ELN gaps are of particular urgency given the system's current use in generating and storing nonclinical data supporting an active IND. The eTMF gaps, while less severe, must be remediated before Phase II site activation. This assessment provides a prioritized remediation roadmap in Section 6. The Company should allocate dedicated resources to the 90-day critical remediation plan immediately and should not expand use of the ELN to additional regulated activities until IQ/OQ/PQ and electronic signature compliance have been achieved.

Section 2 — Regulatory Background

2.1 Overview of 21 CFR Part 11

21 CFR Part 11, published by FDA in March 1997 and clarified by the 2003 Guidance for Industry "Part 11, Electronic Records; Electronic Signatures — Scope and Application," establishes the criteria under which FDA will consider electronic records to be equivalent to paper records and electronic signatures equivalent to traditional handwritten signatures. The rule applies to electronic records that: (a) are created, modified, maintained, archived, retrieved, or transmitted under any FDA-administered records requirement; and (b) are submitted to FDA in electronic format. For clinical-stage biotechnology companies, Part 11 applies to all electronic records generated in the conduct of clinical trials subject to 21 CFR Part 312 (IND regulations), including but not limited to clinical data, safety reports, protocol amendments, site correspondence, and laboratory records.

The 2003 Scope and Application Guidance narrowed the practical scope of Part 11 enforcement by directing FDA resources toward higher-risk systems, emphasizing the importance of predicate rule compliance, and clarifying that the agency would exercise enforcement discretion with respect to certain Part 11 requirements for systems implemented before the rule's effective date. However, this guidance does not constitute a safe harbor from Part 11 obligations, and FDA has consistently cited Part 11 violations in 483 observations and Warning Letters to clinical-stage companies, particularly in the areas of audit trail completeness, electronic signature integrity, and system validation.

The regulatory framework relevant to BioVenture's systems is further informed by: FDA's Data Integrity and Compliance with Drug CGMP Guidance for Industry (December 2018); the MHRA GXP Data Integrity Guidance and Definitions (March 2018, widely adopted as best practice by U.S.-regulated companies); and the FDA Warning Letter to Cetero Research (2010) and subsequent enforcement actions establishing FDA's expectations for audit trail review in clinical settings.

2.2 Application to eTMF Systems

An electronic Trial Master File serves as the authoritative repository for all essential documents that individually and collectively permit evaluation of the conduct of a trial and the quality of the data produced, as defined by ICH E6(R3) Section 8. The eTMF is therefore a primary repository of FDA-regulated electronic records subject to both 21 CFR Part 11 and the predicate rules of 21 CFR § 312.57 (record retention) and ICH E6(R3).

Critical Part 11 requirements for eTMF systems include: validation of the system to ensure accuracy, reliability, and consistent performance (§ 11.10(a)); generation of complete and computer-generated audit trails capturing document creation, modification, deletion, and metadata changes (§ 11.10(e)); implementation of access controls limiting system access to authorized users (§ 11.10(d) and (g)); and use of compliant electronic signatures for document approval workflows, including proper signature manifestations (§ 11.50). The eTMF must also protect records to enable accurate and ready retrieval throughout the required retention period (§ 11.10(c)), which for clinical trial documents extends a minimum of 2 years following the latest marketing application or, if no application is filed, 2 years after the clinical investigation is discontinued and FDA is notified per 21 CFR § 312.62.

2.3 Application to Electronic Laboratory Notebooks

Electronic laboratory notebooks used to record, store, or retrieve nonclinical or clinical laboratory data that support an IND or NDA are subject to 21 CFR Part 11. For BioVenture's ELN, which is used to document in vitro pharmacology, biochemical assay data, and stability measurements supporting IND 145-2204, Part 11 compliance is required under 21 CFR Part 312 (predicate rule) and, for GLP-applicable nonclinical studies, under 21 CFR Part 58.

The ELN context presents specific technical challenges for Part 11 compliance. Laboratory notebooks have historically been paper-based, with handwritten entries providing inherent data integrity assurance through permanence of ink, chronological sequencing, and physical witness signatures. An ELN must replicate these integrity properties electronically: entries must be attributable to the individual who made them (§ 11.10(d)), time-stamped to an accurate system clock (§ 11.10(e)), protected from unauthorized alteration (§ 11.10(a)), and signed by authorized individuals using compliant electronic signatures that carry the same legal weight as handwritten signatures (§§ 11.100, 11.200).

The absence of any of these protections in an ELN creates a data integrity vulnerability that FDA inspectors are specifically trained to identify.

FDA Inspection Focus: ELN Data Integrity

Since 2018, FDA has issued 483 observations and Warning Letters citing ELN-related data integrity failures including: lack of system validation; audit trails that could be disabled by users; shared login credentials enabling non-attributable entries; and electronic signatures that did not bind to the underlying record. BioVenture's ELN gaps identified in this assessment are consistent with the most commonly cited ELN deficiencies in FDA enforcement actions. These are not hypothetical risks.

Section 3 — Scope of Assessment

3.1 Systems Assessed

System	Vendor / Version	Primary Use	GxP Applicable	Part 11 Scope
Veeva Vault eTMF	Veeva Systems / v24.1.0	Storage and management of all Trial Master File documents for BVI-2204	Yes (ICH E6(R3), 21 CFR § 312.57)	Full — all document workflows and electronic signatures
LabArchives ELN	LabArchives LLC / v8.3.2	Recording of in vitro pharmacology, bioanalytical methods, and nonclinical stability data supporting IND 145-2204	Yes (21 CFR Part 58 for GLP studies; 21 CFR Part 312 for IND support)	Full — all laboratory records and electronic signatures

3.2 Assessment Methodology

The assessment was conducted over a period of three weeks using the following methods:

1. Document Review: Review of all system validation documentation, configuration records, SOPs, user access logs, audit trail exports, and vendor qualification documentation made available by BioVenture.
2. System Configuration Testing: Hands-on testing of both systems against a structured test protocol derived from the 21 CFR Part 11 requirements checklist (Appendix A), including testing of audit trail functionality, access control configurations, and electronic signature workflows.
3. Structured Interviews: Approximately 12 hours of structured interviews conducted with: the eTMF System Administrator (1); the ELN System Administrator (1); the Head of Quality Assurance (1); the Director of Regulatory Affairs (1); and a sample of five scientist end-users of the ELN.
4. Audit Trail Review: Export and analysis of 90 days of audit trail data from both systems, reviewed against the completeness requirements of 21 CFR § 11.10(e).
5. Benchmark Review: Assessment of BioVenture's practices against FDA guidance documents, Warning Letter precedents, and industry best practice frameworks including ISPE GAMP 5 (2nd Ed., 2022) and the PIC/S PI 041-1 Data Integrity Guidance.

3.3 Documents Reviewed

Document	Document ID / Version	Status
Veeva Vault eTMF — System Validation Plan	VP-eTMF-001 Rev A	**PASS**
Veeva Vault eTMF — Installation Qualification Report	IQ-eTMF-001 Rev A	**PASS**
Veeva Vault eTMF — Operational Qualification Report	OQ-eTMF-001 Rev A	**PASS**
Veeva Vault eTMF — Performance Qualification Protocol	PQ-eTMF-001 Rev A	**OPEN**
Veeva Vault eTMF — User Access Management SOP	SOP-IT-014 Rev 2	**PASS**
Veeva Vault eTMF — Audit Trail Export (90 days)	Generated Mar 1, 2026	**PASS**
LabArchives ELN — System Configuration Record	No formal document exists	**FAIL**
LabArchives ELN — Validation Documentation (IQ/OQ/PQ)	No documentation exists	**FAIL**

LabArchives ELN — User Access Log	Generated Mar 1, 2026	**PASS**
LabArchives ELN — Electronic Signature Configuration Record	No formal document exists	**FAIL**
BioVenture SOP: Audit Trail Review Procedure	No SOP exists	**FAIL**
BioVenture SOP: Electronic Records Management	SOP-RA-007 Rev 1	**PASS**
Vendor Qualification Package — Veeva Systems	VQP-VEEVA-001	**PASS**
Vendor Qualification Package — LabArchives LLC	No formal package	**FAIL**
FDA 21 CFR Part 11 Certification Letter (§ 11.100(c))	Not submitted to FDA	**FAIL**

Section 4 — Detailed Findings

4a. System Validation — 21 CFR § 11.10(a)

21 CFR § 11.10(a) requires that electronic record systems be validated to ensure accuracy, reliability, consistent intended performance, and the ability to discern invalid or altered records. The ISPE GAMP 5 framework provides the industry-standard approach to computerized system validation (CSV) in pharmaceutical settings, classifying systems by their configuration complexity and directing validation effort commensurate with risk.

Finding 4a-1: eTMF Validation — IQ and OQ Complete; PQ Draft Only (YELLOW)

Veeva Vault eTMF has completed IQ (Installation Qualification Report IQ-eTMF-001, Approved February 2024) and OQ (Operational Qualification Report OQ-eTMF-001, Approved March 2024). The IQ verifies correct system installation with appropriate configuration settings; the OQ tests 187 functional requirements against pre-specified acceptance criteria, with 183 passing and 4 initially failing (subsequently resolved and re-tested). However, the Performance Qualification (PQ), which is required to demonstrate that the system performs consistently under production conditions using actual users and production data, exists only in draft protocol form (PQ-eTMF-001 Rev A, DRAFT). PQ execution has not commenced. The system is currently in production use for BVI-2204 eTMF management despite the absence of a completed PQ. Under GAMP 5 Category 4 software classification applicable to configured commercial software such as Veeva Vault, PQ is required to demonstrate fitness for intended use. Without PQ completion, the validation lifecycle is formally incomplete, and the system cannot be considered fully validated in the regulatory sense.

Specific Gap — PQ Protocol Deficiency

Review of PQ-eTMF-001 Rev A (DRAFT) reveals that the PQ protocol does not include test scripts for: (1) end-to-end document lifecycle workflows reflecting actual BVI-2204 trial document types; (2) user acceptance testing by qualified clinical operations and regulatory personnel; and (3) performance testing under expected production load conditions (multi-site, multi-user concurrent access). These gaps must be addressed before PQ execution.

Finding 4a-2: ELN Validation — No IQ/OQ/PQ Documentation Exists (CRITICAL — RED)

LabArchives ELN has no validation documentation of any kind. No Validation Plan, User Requirements Specification (URS), Design Specification, IQ, OQ, or PQ exists. The system was deployed in October 2023 by the Head of Research without Quality Assurance involvement, based solely on the vendor's published feature list and a 30-day free trial. The system is currently used to record and electronically sign in vitro pharmacology data, biochemical assay results, and analytical chemistry records, all of which support IND 145-2204 and are subject to 21 CFR Part 11.

Interviews with ELN users revealed that: (a) users are aware that the system is not "fully set up" for regulatory use but continue to use it pending IT's validation effort; (b) four users share a single "lab administrator" account to upload batch data files, creating a non-attributability problem; and (c) the system clock synchronization source has not been verified against a traceable time standard, meaning timestamps on laboratory records cannot be confirmed to be accurate. These findings collectively represent a serious data integrity failure under 21 CFR § 11.10(a) and will be viewed by FDA inspectors as a systemic quality management failure, not merely a technical gap.

Remediation Urgency

The ELN must be placed in a formal validation hold — meaning no new GxP records should be created in the system and no ELN records should be referenced in IND amendments or safety reports — until IQ/OQ/PQ is completed. Existing records must be subject to a data integrity assessment by QA to determine whether they retain evidentiary value. This is a critical remediation action.

4b. Audit Trails — 21 CFR § 11.10(e)

21 CFR § 11.10(e) requires that systems used to maintain records subject to FDA regulation generate secure, computer-generated, time-stamped audit trails that independently record the date and time of operator entries and actions that create, modify, or delete electronic records. The audit trail must record who made the change, when they made it, and what was changed, including the original value of any modified field. Crucially, audit trails must be protected

against modification and must be available for review by FDA inspectors. The FDA's 2018 Data Integrity Guidance further requires that audit trails be reviewed periodically by qualified personnel as part of routine quality oversight.

Finding 4b-1: eTMF Audit Trail — Technically Complete; Review SOP Absent (YELLOW)

Veeva Vault's audit trail functionality is enabled and properly configured. Review of 90 days of exported audit trail data confirms that the system captures: document creation, modification, deletion, and restoration events; user identity (authenticated username); timestamp (UTC, synchronized to NTP); action type; field-level changes with old and new values; and document version changes. These elements satisfy the technical requirements of § 11.10(e). Audit trail data is stored in a separate, access-controlled database partition and cannot be modified by any user, including system administrators, satisfying the protection requirement.

However, no SOP exists for periodic audit trail review, and no audit trail reviews have been performed since the system went live in September 2023. This is a governance gap, not a technical gap — but FDA's 2018 Data Integrity Guidance is explicit that audit trails must be reviewed with appropriate frequency, and inspection experience confirms that systems with no audit trail review history will receive 483 observations. The review frequency should be commensurate with the volume of activity and risk: for a Phase II eTMF, monthly review of active document workflows and quarterly comprehensive review is appropriate.

Finding 4b-2: ELN Audit Trail — Partial Coverage; Critical Gap in Metadata Capture (YELLOW)

LabArchives ELN generates audit trail records for notebook entry creation and text editing events. However, the following significant gaps were identified:

- File attachment modifications are not captured in the audit trail. In laboratory practice, attachments (instrument output files, chromatograms, spectral data) constitute primary raw data. Changes to attachments are not visible in the ELN audit trail, creating a potential data integrity gap for regulatory submissions relying on instrument data.
- System administrator actions — including user permission changes, notebook visibility changes, and account creation/deactivation — are not captured in the ELN audit trail. This means that administrative manipulation of the record environment cannot be detected through audit trail review, a gap that FDA considers a critical data integrity risk.
- The audit trail for deleted entries captures that a deletion occurred but does not capture the content of the deleted record (original value). This fails the § 11.10(e) requirement that audit trails capture the original value of modified or deleted data.
- System clock synchronization has not been verified. ELN timestamps cannot be confirmed to derive from a traceable time source, raising potential evidentiary concerns about the accuracy of dates and times recorded in laboratory records.

4c. Access Controls — 21 CFR § 11.10(d) and (g)

21 CFR §§ 11.10(d) and 11.10(g) require that electronic record systems: limit system access to authorized individuals (§ 11.10(d)); use operational system checks to enforce permitted sequencing of steps and events (§ 11.10(f)); use authority checks to ensure that only authorized individuals can use the system, electronically sign a record, access the operation or computer system input or output device, alter a record, or perform the operation at issue (§ 11.10(g)). These requirements collectively constitute a role-based access control framework anchored to individual user identity.

Finding 4c-1: eTMF Access Controls — Compliant; Periodic Access Review Needs Formalization (GREEN/YELLOW)

Veeva Vault eTMF access controls are substantially compliant. The system implements individual user accounts with unique usernames and passwords; role-based access control (RBAC) with seven configured roles mapped to job functions (Study Coordinator, Site Monitor, Regulatory Affairs, QA, Read-Only Reviewer, eTMF Administrator, System Administrator); multi-factor authentication (MFA) enabled for all users; and automatic session timeout after 15 minutes of inactivity. Password complexity requirements (minimum 12 characters, complexity rules, 90-day rotation) are enforced by the system.

SOP-IT-014 (User Access Management) governs account provisioning and de-provisioning. Review of the access log indicates that four former contractors who left the project in Q4 2024 still have active accounts in the system, suggesting the de-provisioning process is not consistently followed. While these accounts have not been accessed since the contractors' departure (confirmed by access log), their continued existence represents a control weakness. Additionally, no formal periodic access review (commonly known as a "user access review" or UAR) has been conducted since system go-live. SOP-IT-014 requires quarterly UARs but none has been documented.

Finding 4c-2: ELN Access Controls — Shared Account Use; Materially Non-Compliant (YELLOW)

LabArchives ELN access controls present two significant compliance gaps identified through interviews and access log review:

- Shared Account Use: Four members of the analytical chemistry team share a single "LabAdmin" account to upload batch analytical data files. This practice directly violates 21 CFR § 11.10(d)'s requirement that system access be limited to authorized individuals (implying individual, attributable accounts). Records created or modified under the shared account cannot be attributed to an individual scientist, violating the data integrity principle of attributability (the "A" in ALCOA+). The scientists report that the shared account was established because individual account uploads were "too slow" for large data files — a performance limitation that must be addressed through technical remediation, not by abandoning access control requirements.
- Absence of Formal Role-Based Access Controls: LabArchives has been configured with only two access levels: "Owner" (full access, including ability to delete entries) and "Member" (read/write access). No separation of duties controls prevent a scientist from both creating a record and approving it. No restricted "read only" role exists for external reviewers or auditors. This configuration is inconsistent with the minimum necessary access principle and the authority check requirements of § 11.10(g).

4d. Electronic Signatures — 21 CFR §§ 11.50, 11.100, 11.200

21 CFR Part 11 Subpart C establishes requirements for electronic signatures that are to be considered legally equivalent to handwritten signatures. The core requirements are: (1) each electronic signature must be unique to one individual and not reused or reassigned (§ 11.100(a)); (2) the identity of the individual signing must be verified before the electronic signature is established (§ 11.100(b)); (3) persons using electronic signatures must certify to FDA that their electronic signatures are legally binding equivalents of their handwritten signatures (§ 11.100(c)); (4) electronic signatures must employ at least two distinct identification components (e.g., an identification code and a password) for signatures not based on biometrics (§ 11.200(a)(1)); (5) the signature manifestation must display the printed name of the signer, the date and time of signing, and the meaning of the signature (e.g., review, approval, responsibility, authorship) (§ 11.50(a)).

Finding 4d-1: eTMF Electronic Signatures — Partial Compliance; Manifestation Gap for Amendment Approvals (YELLOW)

Veeva Vault eTMF implements electronic signatures using individual username/password combinations (satisfying the two-component requirement of § 11.200(a)(1)), with signatures uniquely linked to each user's authenticated account (satisfying § 11.100(a)). User identity verification was conducted during account setup (satisfying § 11.100(b)). The signature manifestation for initial document approval workflows correctly displays the printed name, timestamp, and signature meaning ("Approved").

However, two specific non-compliances were identified:

- Signature Manifestation for Amendment/Revision Approvals: When a document is revised and re-approved after an initial approval, the system displays the revision approval signature in metadata only (accessible through the audit trail) and does not display it visibly on the face of the document. For the revision approval to satisfy § 11.50(a), the manifestation must be visible on the signed document — not only in the audit trail. This is a configuration gap in the Vault lifecycle settings that can be remediated without vendor engagement.
- FDA Certification Not Submitted: BioVenture has not submitted the § 11.100(c) certification to FDA stating that electronic signatures used in electronic records submitted to the agency are intended to be the legally binding equivalents of traditional handwritten signatures. This certification is a mandatory prerequisite for use of electronic signatures in records submitted to FDA.

Finding 4d-2: ELN Electronic Signatures — Non-Compliant on Multiple Dimensions (CRITICAL — RED)

LabArchives ELN's electronic signature implementation is materially non-compliant with 21 CFR Part 11 Subpart C requirements:

- Single-Component Signature: The ELN's "sign" function requires only username entry (no re-entry of password or second identification component at the time of signing). This violates § 11.200(a)(1)'s requirement for at least two distinct identification components. A user who leaves a workstation without logging out could have records signed under their credentials by another person without the second authentication barrier.
- Signature Manifestation Non-Compliant: The signature block visible on ELN entries displays only the username and a timestamp. It does not display: (a) the printed full name of the signer (§ 11.50(a)(1)); or (b) the meaning of the signature (e.g., "Author," "Reviewer," "Approved") (§ 11.50(a)(3)). The printed name requirement is not satisfied by a username abbreviation.
- Signature-Record Linkage Not Demonstrated: The ELN has not been configured or validated to demonstrate that electronic signatures are linked to their respective electronic records such that signatures cannot be excised, copied, or otherwise transferred to falsify another record. This requirement of § 11.70 has not been evaluated or documented.
- No § 11.100(c) Certification: As noted for the eTMF, BioVenture has not submitted the required FDA certification for electronic signatures generally.

4e. Records Retention and Recovery — 21 CFR §§ 11.10(b), (c)

21 CFR §§ 11.10(b) and (c) require that systems generate accurate and complete copies of electronic records in both human-readable and electronic form suitable for inspection, review, and copying by FDA (§ 11.10(b)), and protect records to enable their accurate and ready retrieval throughout the required retention period (§ 11.10(c)). For clinical trial records under 21 CFR § 312.62, the retention period is at minimum 2 years after the latest marketing application or 2 years after the investigation is discontinued. In practice, BioVenture should plan for a 15-year retention period to cover a potential NDA filing and post-approval requirements.

Finding 4e-1: eTMF Records Retention — Compliant; Disaster Recovery Test Not Documented (GREEN/YELLOW)

Veeva Vault eTMF stores all records in Veeva's SOC 2 Type II-certified cloud infrastructure with geographic redundancy (primary: US-East-1; secondary: US-West-2), daily automated backups with 365-day retention, and 99.9% uptime SLA. The system generates human-readable PDFs of all documents with embedded metadata (including audit trail references) satisfying § 11.10(b). Veeva's Data Retention configuration has been set to the 15-year retention period recommended by QA.

One gap was identified: BioVenture has not conducted a documented disaster recovery (DR) test for the eTMF system. While Veeva conducts infrastructure DR testing, BioVenture's own SOP for business continuity does not include an eTMF-specific DR exercise and has not verified that records can be accurately retrieved from backup in a reasonable timeframe. This is a governance gap that presents low immediate risk (given Veeva's infrastructure reliability) but should be remediated to demonstrate control over the system's long-term integrity.

Finding 4e-2: ELN Records Retention — Partial Compliance; No Formal Backup Verification (YELLOW)

LabArchives provides cloud-hosted storage with daily backups as part of its standard service. However:

- No formal retention configuration has been established for BioVenture's ELN account. The system default allows notebook owners to delete entries permanently. This capability has not been restricted, meaning laboratory records could be permanently deleted by any user with Owner-level access — a direct violation of records protection requirements under § 11.10(c).
- No backup and recovery verification has been conducted. BioVenture cannot confirm that ELN records can be accurately and completely recovered in the event of data loss, system failure, or account compromise.
- No Vendor Qualification Package exists for LabArchives. BioVenture has not obtained or evaluated LabArchives's SOC 2 report, uptime history, data retention policies, or breach notification procedures. The vendor's obligations regarding data retention, ownership, and return upon contract termination have not been documented in a Data Processing Agreement meeting the requirements of data protection regulations.

Section 5 — Gap Analysis Table

The following gap analysis summarizes all findings identified in Section 4, with risk ratings, recommended remediations, and timelines. Risk ratings reflect the potential impact on regulatory compliance, data integrity, and FDA inspection outcome: CRITICAL (potential FDA Warning Letter or Import Alert; immediate patient safety or data integrity risk); HIGH (likely 483 observation; significant compliance remediation required); MEDIUM (483 observation risk; process improvement required); LOW (best practice gap; low inspection risk).

Gap / Finding	Current State	Regulatory Impact	Risk	Recommended Remediation	Timeline
ELN: No validation documentation (IQ/OQ/PQ) exists for LabArchives	No URS, IQ, OQ, or PQ documentation exists; system deployed without QA involvement	System operates entirely outside validated state; all ELN records lack regulatory integrity guarantee	**CRITICAL**	Immediately place ELN in validation hold. Engage GAMP 5-experienced CSV consultant. Develop URS, IQ, OQ, PQ per LabArchives GAMP 5 Category 4 framework. Target: IQ+OQ complete within 60 days; PQ complete within 90 days.	**0–90 days**
ELN: Electronic signatures non-compliant — single component; manifestation errors; no § 11.70 linkage validation	Single-password signing; no printed name or meaning in manifestation; signature-record linkage undemonstrated	ELN electronic signatures are not legally valid for FDA submissions; signed records may not constitute binding approvals	**CRITICAL**	Reconfigure ELN signature workflow to require username + password at signing. Update manifestation template to include printed name and meaning. Validate signature-record linkage as part of OQ. Complete before any ELN records are submitted to FDA.	**0–90 days**
Both systems: FDA § 11.100(c) certification letter not submitted	No certification submitted to FDA	Electronic signatures technically not established as legally binding equivalents of handwritten signatures under federal law	**CRITICAL**	Draft and submit § 11.100(c) certification to FDA within 30 days. Have General Counsel review and execute. File with FDA Dockets Management and retain copy in regulatory files.	**0–30 days**
ELN: Shared "LabAdmin" account used by four scientists for data uploads	Four individuals share one account for batch data file uploads	Records uploaded under shared account are non-attributable; violates § 11.10(d) and ALCOA+ attributability principle	**HIGH**	Immediately cease shared account use. Create individual accounts for all four scientists. Configure bulk upload capability for individual accounts. Retroactively assess existing shared-account records for data integrity impact.	**0–30 days**
eTMF: Performance Qualification (PQ) not completed; system in production use	PQ protocol exists as draft; no PQ execution has commenced	eTMF validation lifecycle formally incomplete; system technically not fully validated under GAMP	**HIGH**	Finalize PQ protocol to include BVI-2204-specific document type workflows and production user acceptance testing. Execute PQ and resolve all deviations. Target PQ completion before Phase II site activation.	**30–60 days**

		5			
eTMF: Signature manifestation for amendment/revision approvals not displayed on document face	Revision approval signatures visible only in audit trail metadata, not on document face	§ 11.50(a) requires manifestation on signed document; audit-trail-only visibility does not satisfy requirement	**HIGH**	Update Veeva Vault lifecycle configuration for all document types to display revision approval signatures on the document face (visible in PDF rendition). Validate change through OQ re-test of affected workflows.	**30–60 days**
Both systems: No audit trail review SOP; no reviews conducted since go-live	No SOP governs audit trail review; no review has been performed in 6+ months of operation	FDA Data Integrity Guidance (2018) requires periodic audit trail review; absence is a commonly cited 483 observation	**HIGH**	Draft and approve SOP for periodic audit trail review (cover both eTMF and ELN). Define review frequency (monthly active workflow review; quarterly comprehensive review), reviewer qualifications, and documentation requirements. Conduct catch-up review for past 6 months.	**30–60 days**
ELN: Administrator actions not captured in audit trail; attachment modifications not logged	Audit trail does not capture admin actions or file attachment changes	Material data integrity events (admin permission changes, raw data file alterations) are invisible to audit review	**HIGH**	Escalate to LabArchives vendor: request audit trail enhancement for admin actions and attachment events, or evaluate alternative ELN platform with complete audit trail coverage. Document gap in risk assessment pending vendor response.	**30–90 days**
ELN: Deleted record content not captured in audit trail (original value missing)	Audit trail records deletion event but not the content of the deleted record	Violates § 11.10(e) requirement to capture original value; prevents reconstruction of deleted records	**HIGH**	Configure ELN to capture deleted record content in audit trail prior to deletion, or implement workflow requiring QA approval before any entry can be deleted. Evaluate vendor capability and document workaround if not available natively.	**30–90 days**
eTMF: Four former contractor accounts not deactivated	Access log shows four departed contractors retain active accounts	Unnecessary access privileges; control weakness in de-provisioning process; regulatory inspection finding	**MEDIUM**	Immediately deactivate the four identified accounts. Conduct full user access review against current authorized personnel list. Revise SOP-IT-014 to require de-provisioning within 24 hours of contractor departure.	**0–30 days**
eTMF: Quarterly user access review not conducted per SOP	SOP-IT-014 requires quarterly UAR; none documented since go-live	Unauthorized or inappropriate access may persist; control not operating as designed	**MEDIUM**	Conduct overdue UAR immediately. Document results. Establish recurring calendar reminder for subsequent quarterly UARs. Assign responsibility to eTMF System Administrator with QA oversight.	**30–60 days**

ELN: Only two access roles configured (Owner/Member); no read-only or restricted role	Insufficient role granularity; all active users have write access; no separation between creation and approval	Separation of duties not enforced; single user could create and approve records without independent review	**MEDIUM**	Configure additional ELN roles: Read-Only (for auditors and external reviewers), Reviewer (read/comment only, no editing), and Author (create/edit, cannot approve). Assign all users to appropriate roles.	**30–90 days**
ELN: No vendor qualification package for LabArchives	No SOC 2 report review, uptime assessment, or Data Processing Agreement in place	Vendor reliability and data protection obligations are unverified; data loss risk cannot be assessed	**MEDIUM**	Obtain LabArchives SOC 2 Type II report (current year). Execute Data Processing Agreement. Assess backup and recovery capabilities. Document vendor qualification in VQP-LABARCHI-001.	**30–90 days**
ELN: Permanent record deletion capability available to all Owner-level users	ELN default allows permanent deletion by notebook owners	Records may be permanently deleted without detection; violates § 11.10(c) protection requirement	**MEDIUM**	Configure ELN to disable permanent deletion for all non-system-administrator accounts. Implement soft-delete workflow requiring QA review before any entry removal. Validate configuration.	**30–90 days**
ELN: System clock synchronization to traceable time source not verified	No documentation of NTP synchronization or time accuracy verification	ELN timestamps may not be accurate; evidentiary value of time-stamped records is uncertain	**MEDIUM**	Document ELN system clock synchronization configuration. Verify against traceable time standard (e.g., NIST time server). Test timestamp accuracy as part of OQ protocol. Document in IQ.	**60–90 days**
eTMF: No disaster recovery test documented for eTMF-specific data	Business continuity SOP does not include eTMF DR test; Veeva infrastructure DR assumed adequate	DR capability for eTMF records not verified at application layer; long-term retention risk	**LOW**	Add eTMF-specific DR test to annual business continuity exercise plan. Conduct table-top exercise to verify record retrieval from backup. Document results.	**90–180 days**

Section 6 — Remediation Roadmap

6.1 90-Day Critical Remediation Plan

The following sprint plan addresses all CRITICAL and HIGH findings identified in this assessment. Successful execution of this plan will achieve validated status for the ELN, full 21 CFR Part 11 compliance for electronic signatures in both systems, and eTMF PQ completion — the three minimum requirements before BVI-2204 Phase II site activation. Resource owners are identified by functional role; BioVenture should identify specific named individuals responsible for each action within one week of receipt of this report.

Timeline	Priority	Action	Owner	Success Criterion
Week 1–2	**CRITICAL**	Submit 21 CFR § 11.100(c) certification letter to FDA	Regulatory Affairs / General Counsel	Executed certification letter on FDA docket
Week 1–2	**CRITICAL**	Cease use of shared "LabAdmin" ELN account; create individual accounts for all four scientists	IT / ELN System Administrator	Zero shared-account logins; individual accounts verified in access log
Week 1–2	**CRITICAL**	Initiate formal ELN validation hold: QA memo to all ELN users; no new GxP records pending validation completion	QA / Head of Research	QA hold memo issued; ELN use restricted to non-GxP activities
Week 1–2	**HIGH**	Deactivate four former contractor accounts in eTMF; conduct immediate user access review	IT / eTMF Administrator	Zero inactive accounts; UAR documentation complete
Week 2–4	**CRITICAL**	Engage GAMP 5 CSV consultant; initiate ELN validation — draft URS and Validation Plan	QA / External CSV Consultant	Approved URS and Validation Plan; IQ protocol in draft
Week 2–4	**HIGH**	Draft and issue SOP for audit trail review (both eTMF and ELN)	QA / Regulatory Affairs	SOP approved through QMS; first review scheduled
Week 2–4	**HIGH**	Update eTMF lifecycle configuration: revision approval signature on document face	eTMF Administrator / IT	Configuration change implemented; OQ re-test passed
Week 3–6	**CRITICAL**	Reconfigure ELN signature workflow: two-component authentication; updated manifestation template	ELN Administrator / IT	Reconfigured; signature validation test protocol passed
Week 4–8	**HIGH**	Execute eTMF PQ: finalize protocol; conduct user acceptance testing with clinical operations staff	QA / Clinical Operations	Approved PQ Report; all critical test cases passed
Week 6–10	**CRITICAL**	Complete ELN IQ and OQ; address audit trail gaps; validate system clock synchronization	QA / External CSV Consultant	Approved IQ and OQ Reports; all critical OQ test cases passed; Part 11 configuration verified
Week 8–12	**CRITICAL**	Conduct ELN PQ with production users; conduct data integrity retrospective assessment of pre-validation ELN records	QA / Research / External Consultant	Approved PQ Report; retrospective data integrity assessment report issued
Week 10–12	**HIGH**	Configure ELN role-based access controls (Author, Reviewer, Read-Only); disable permanent deletion capability	ELN Administrator	Role assignments complete; deletion restriction validated

6.2 12-Month Full Compliance Plan

The 12-month plan extends the 90-day sprint to address MEDIUM and LOW findings, establish ongoing governance processes (periodic audit trail review, quarterly user access reviews, annual Part 11 compliance reviews), and achieve FDA inspection readiness for both systems. The plan culminates in a mock FDA inspection readiness review at Month 10, providing BioVenture with an independent assessment of inspection preparedness before any FDA inspection of BVI-2204 is anticipated.

Milestone	Risk Level	Activity	Owner	Deliverable
Months 1–3	**CRITICAL (all)**	Complete all 90-day critical remediations above	Per above	All CRITICAL findings closed; eTMF fully validated; ELN validated and Part 11-compliant
Month 2	**HIGH**	Obtain LabArchives SOC 2 Type II report; execute Data Processing Agreement; complete Vendor Qualification Package	Legal / IT / QA	VQP-LABARCHI-001 approved; DPA executed
Month 2	**MEDIUM**	Conduct first formal quarterly eTMF user access review per SOP-IT-014	eTMF Administrator / QA	Documented UAR; all findings remediated
Month 3	**HIGH**	Issue retrospective data integrity assessment report for pre-validation ELN records; determine regulatory usability of existing records	QA / External Consultant	Written report; regulatory strategy determination for each record category
Month 4	**MEDIUM**	Conduct first periodic audit trail review for both eTMF and ELN under new SOP	QA Reviewer	Documented audit trail review reports; findings addressed
Month 4–5	**MEDIUM**	Implement ELN backup and recovery verification; conduct first DR test	IT / QA	DR test report; recovery time and data completeness verified
Month 5	**MEDIUM**	Develop and approve 21 CFR Part 11 training module; deploy to all system users	QA / Training / Regulatory	100% user training completion; LMS records updated
Month 6	**ALL**	Conduct internal 6-month compliance review: assess status of all findings; verify closure evidence	QA / Regulatory Affairs	6-month compliance review report; updated findings register
Month 7–9	**MEDIUM / LOW**	Address any residual medium and low findings; refine SOPs based on operational experience	QA / System Owners	All MEDIUM findings closed; LOW findings on remediation plan
Month 10	**ALL**	Prepare for mock FDA inspection readiness review: assemble validation packages, audit trail exports, access control records, SOP library	QA / Regulatory Affairs / External Consultant	Mock inspection completed; readiness report issued; gaps closed
Month 12	**ALL**	Annual 21 CFR Part 11 compliance review: assess both systems against current regulatory requirements; update gap register	QA / Regulatory Affairs / External Consultant	Annual compliance review report; updated risk register; 12-month forward remediation plan

Resource Recommendation

BioVenture should engage a GAMP 5-experienced external CSV consultant to lead ELN validation (estimated 3–4

months, ~$80,000–$120,000 depending on ELN platform complexity and extent of existing configuration documentation). Internal QA capacity should be assessed: the current QA team of two is likely insufficient to execute this remediation plan alongside routine trial quality oversight. A contract QA specialist with 21 CFR Part 11 and ELN validation expertise should be engaged for the 90-day critical remediation period.

Appendix A — 21 CFR Part 11 Requirements Compliance Checklist

The following checklist maps each 21 CFR Part 11 regulatory requirement to the observed compliance status of each system assessed. Compliance status reflects the state observed during this assessment and is subject to change as remediation actions are implemented. "PARTIAL" indicates that the requirement is partially implemented but has identified gaps. This checklist should be updated by BioVenture's QA team upon completion of each remediation action and retained as part of the validation documentation package.

Regulation	Requirement Summary	eTMF Status	ELN Status
§ 11.10(a)	Systems used to create, modify, maintain, or transmit electronic records shall be validated to ensure accuracy, reliability, consistent intended performance, and ability to discern invalid or altered records.	**PARTIAL**	**NON-COMPLIANT**
§ 11.10(b)	The ability to generate accurate and complete copies of records in both human readable and electronic form suitable for inspection, review, and copying by the agency.	**COMPLIANT**	**PARTIAL**
§ 11.10(c)	Protection of records to enable their accurate and ready retrieval throughout the records retention period.	**COMPLIANT**	**PARTIAL**
§ 11.10(d)	Limiting system access to authorized individuals.	**PARTIAL**	**NON-COMPLIANT**
§ 11.10(e)	Use of secure, computer-generated, time-stamped audit trails to independently record the date and time of operator entries and actions that create, modify, or delete electronic records.	**COMPLIANT**	**PARTIAL**
§ 11.10(f)	Use of operational system checks to enforce permitted sequencing of steps and events, as appropriate.	**COMPLIANT**	**PARTIAL**
§ 11.10(g)	Use of authority checks to ensure that only authorized individuals can use the system, electronically sign a record, access the computer system, alter a record, or perform the operation at issue.	**COMPLIANT**	**PARTIAL**
§ 11.10(h)	Use of device (e.g., terminal) checks to determine, as appropriate, the validity of the source of data input or operational instruction.	**COMPLIANT**	**COMPLIANT**
§ 11.10(i)	Determination that persons who develop, maintain, or use electronic record/electronic signature systems have the education, training, and experience to perform their assigned tasks.	**PARTIAL**	**PARTIAL**
§ 11.10(j)	The establishment of, and adherence to, written policies that hold individuals accountable for actions initiated under their electronic signatures.	**PARTIAL**	**NON-COMPLIANT**
§ 11.10(k)	Use of appropriate controls over systems documentation including: adequate controls over the distribution of, access to, and use of documentation for system operation and maintenance.	**COMPLIANT**	**PARTIAL**
§ 11.30	Controls for open systems: persons who use open systems to create, modify, maintain, or transmit electronic records shall employ procedures and controls designed to ensure authenticity, integrity, and confidentiality.	**N/A**	**N/A**
§ 11.50(a)(1)	Signed electronic records shall contain information associated with the signing that clearly indicates the printed name of the signer.	**COMPLIANT**	**NON-COMPLIANT**
§ 11.50(a)	The date and time when the signature was executed.	**COMPLIANT**	**PARTIAL**

(2)			
§ 11.50(a)(3)	The meaning (such as review, approval, responsibility, or authorship) associated with the signature.	**PARTIAL**	**NON-COMPLIANT**
§ 11.50(b)	The items identified in § 11.50(a)(1)–(a)(3) shall be subject to the same controls as for electronic records and shall be included as part of any human readable form of the electronic record.	**PARTIAL**	**NON-COMPLIANT**
§ 11.70	Electronic signatures and handwritten signatures executed to electronic records shall be linked to their respective electronic records to ensure that such signatures cannot be excised, copied, or otherwise transferred to falsify an electronic record by ordinary means.	**COMPLIANT**	**NON-COMPLIANT**
§ 11.100(a)	Each electronic signature shall be unique to one individual and shall not be reused by, or reassigned to, anyone else.	**COMPLIANT**	**PARTIAL**
§ 11.100(b)	Before an organization establishes, assigns, certifies, or otherwise sanctions an individual's electronic signature, or any element of such electronic signature, the organization shall verify the identity of the individual.	**COMPLIANT**	**PARTIAL**
§ 11.100(c)	Persons using electronic signatures shall, prior to or at the time of such use, certify to the agency that the electronic signatures in their system, used on or after August 20, 1997, are intended to be the legally binding equivalent of traditional handwritten signatures.	**NON-COMPLIANT**	**NON-COMPLIANT**
§ 11.200(a)(1)	Electronic signatures not based upon biometrics shall employ at least two distinct identification components such as an identification code and password.	**COMPLIANT**	**NON-COMPLIANT**
§ 11.200(a)(2)	When an individual executes a series of signings during a single, continuous period of controlled system access, the first signing shall be executed using all electronic signature components; subsequent signings shall be executed using at least one electronic signature component that is only executable by, and designed to be used only by, the individual.	**COMPLIANT**	**PARTIAL**
§ 11.200(a)(3)	When an individual executes one or more signings not performed during a single period of controlled system access, each signing shall be executed using all of the electronic signature components.	**COMPLIANT**	**NON-COMPLIANT**
§ 11.200(b)	Electronic signatures not based upon biometrics shall be used only by their genuine owners.	**COMPLIANT**	**PARTIAL**
§ 11.300(a)	Maintain the uniqueness of each combined identification code and password, such that no two individuals have the same combination.	**COMPLIANT**	**NON-COMPLIANT**
§ 11.300(b)	Ensure that identification code and password issuances are periodically checked, recalled, or revised.	**PARTIAL**	**NON-COMPLIANT**
§ 11.300(c)	Follow loss management procedures to electronically deauthorize lost, stolen, missing, or otherwise potentially compromised tokens, cards, and other devices that bear or generate identification code or password information.	**COMPLIANT**	**PARTIAL**
§ 11.300(d)	Use transaction safeguards to prevent unauthorized use of passwords and/or identification codes, and to detect and report in an immediate and urgent manner	**COMPLIANT**	**PARTIAL**

	any attempts at their unauthorized use.		
§ 11.300(e)	Initial and periodic testing of devices, such as tokens or cards, that bear or generate identification code or password information to ensure that they function properly and have not been altered in an unauthorized manner.	**N/A**	**N/A**

Legend	Definition
COMPLIANT	Requirement fully met based on system configuration review, documentation audit, and functional testing.
PARTIAL	Requirement partially met; specific gaps identified in Section 4 and Section 5 of this report.
NON-COMPLIANT	Requirement not met; material gap exists requiring remediation before system use for FDA-regulated activities.
N/A	Requirement not applicable to this system type (e.g., § 11.30 open system controls apply only to Internet-facing systems without encryption; both BioVenture systems are closed systems under § 11.3(b)(4)).

Consultant Attestation: This assessment was conducted by Dr. M. Chen (PhD Biomedical Engineering; JD) in accordance with professional regulatory consulting standards. The findings, gap analysis, and recommendations represent the consultant's professional judgment based on the documents reviewed, interviews conducted, and functional testing performed as described herein. This report is prepared as of the assessment date noted on the cover page. Regulatory requirements, system configurations, and BioVenture's compliance posture may change after this date. BioVenture Inc. is responsible for ensuring ongoing compliance with 21 CFR Part 11 and all applicable predicate rules. This report does not constitute legal advice; specific legal questions regarding FDA enforcement posture should be directed to qualified regulatory counsel.